MILITARY SPECIFICATION

CLOTH, BROADCLOTH: WOOL AND WOOL/NYLON

This specification is approved for use by U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for wool and wool/nylon broadcloth fabrics.

1.2 Classification. The broadcloth shall be of the following types as specified (see 6.2):

Type I - Wool
Type II - Wool/Nylon

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth, Wool: Mothproofing Treatment of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps, Headquarters U.S. Marine Corps (Code LMA-1), Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool Worsted and Wool Blends

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS
RULES AND REGULATIONS UNDER THE WOOL PRODUCT LABELING ACT OF 1939

(Application for copy should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

U. S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grade of Wool

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Marketing Service, Washington, D. C. 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

*3.2 First article. When specified (see 4.3, 6.2 and 6.4), the contractor shall furnish a sample of cloth for first article inspection and approval. The first article shall consist of 5 yards, full width. The approval of the first article authorizes the commencement of production, but does not relieve the contractor of the responsibility for compliance with all provisions of the specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production cloth.

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3.3 Material.

3.3.1 Fiber.

* 3.3.1.1 Wool. The wool stock shall be of fleece or pulled wool, or a combination of both, not lower in grade than 60's U.S. Standard. The use of 30% noils is permitted. The use of reprocessed wools, reused wools, or any wool waste, or wool manufacturing by-products, other than those resulting from the supplier's own current production of this cloth, is prohibited. Inspections shall be performed as specified in 4.4.1.1 and 4.4.2.

3.3.2 Wool content.

3.3.2.1 Type I. The finished cloth shall contain not less than 95 percent wool, based on the dry chloroform - extracted specimen, when tested as specified in 4.5.

* 3.3.2.2 Type II. The finished cloth shall be wool blended with first quality nylon fiber in such proportion that the finished cloth shall contain not less than 75 percent wool and not more than 25 percent of the nylon fiber when tested as specified in 4.5. Content shall be determined based on the dry chloroform - extracted specimen.

* 3.3.3 Yarn. The wool and wool nylon fibers shall be blended, carded, and evenly spun on the woolen system into singles yarn for both the warp and the filling when tested as specified in 4.4.1 and 4.5.

* 3.3.4 Color. The color shall be scarlet shade 2501 or dark blue 2307 as specified (see 6.2), and shall be produced by stock or piece dying with suitable dyes to match the standard sample.

* 3.3.4.1 Matching. The finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7000° +500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850° +100° Kelvin.

* 3.3.4.2 Colorfastness. The finished cloth shall show colorfastness to light, wet-dry cleaning, crocking, and perspiration equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the finished cloth shall show "good" fastness to light, wet-dry cleaning, and perspiration and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.5.

3.4 Design and construction.

* 3.4.1 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

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Table I. Physical requirements

Width (Exclusive of selvage) 1/ (Min)		Weight oz/Lin yd 2/		Yarns/inch (min)		Breaking strength (lbs) Min lbs		Shrinkage (percent) (max)		Weave
	Inches	Min	Max	Warp	Filling	Warp	Filling	Warp	Filling	
Type I	56	14	15.5	50	40	50	40	5.0	3.0	2/1
Type II	60	20	22	55	55	83	80	5.0	2.0	2/1

1/ Or tuck in/or leno type weave when woven on shuttleless looms.
2/ Based on specified width.

*3.4.2 Weave. The weave for all types shall be 2 X 1 right hand twill when tested as specified in 4.5.

3.4.3 Selvage yarns. To prevent edges of the cloth from building up when rolled, the single yarn used in the selvage may be made of a finer count than those used in the body of the cloth. The fiber content of the selvage yarns shall be the same as that of the body of the cloth.

3.5 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length of cloth furnished shall be put-up on rolls with the face side inward as specified in PPP-P-1132.

3.6 Finish. The fabric shall have a finish and hand equal to or better than the standard sample. The fabric shall be thoroughly scoured, properly fulled, free from vegetable matter (carbonized if necessary) and the face shall be evenly sheared and pressed.

3.6.1 pH. The pH value of the water extract of the finished cloth shall be not less than 4.0 nor more than 8.0 when tested as specified in 4.5.

*3.6.2 Mothproofing. The cloth shall be given a mothproofing treatment in accordance with MIL-C-43665 (see 4.5).

3.7 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than 1/2 inch in height. The marking shall be clearly legible in any indelible marking used commercially and shall not show through on the face of the cloth.

3.7.1 Wool content label. Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939.

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3.8 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4, shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1-1/2-inch-long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect (see 4.4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).

2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2.1), it shall be examined for appearance, color and finish defects and shall be tested for the characteristics specified in Table II. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-655, except where otherwise indicated herein.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be visually examined and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, a certificate of compliance shall be submitted for the blending and spinning of the yarns as specified in 3.3.3.

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4.4.1.1 Wool. The wool shall be visually examined for grade in scoured form, prior to dyeing, picking, or carding, by comparison with the Methods of Test for Grades of Topwool. The examination for grade shall be performed on a composite sample of 1 pound of scoured wool for each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot. In the event of dispute resulting from comparison, the grade shall be determined by a width method (wedge). The lot shall be unacceptable if the sample does not conform to the requirements of 3.3.1.1. Each lot shall consist of one grade of wool only.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved components are used (see 3.3.1.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.4.3 Examination of the end item. Examination of the end item for yard-by-yard, length, back marking, compliance with the Wool Product Labeling Act, and for defect marking (stringing of defects) shall be as specified in MIL-STD-655. The fabric quality for the finished cloth shall be quality level 10.

* 4.4.3.1 Limits of slubs and knots. Only slubs and knots which exceed "C" for slubs and "D" for knots on the Sears Fabric Defect Scale (see 6.5) shall be scored.

4.4.3.2 Examination for shade and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.3.4), and finish (see 3.6). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish as applicable. The sample unit shall be a 4-inch by 20-inch swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

* 4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3-1/2 yards, full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample sizes shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test methods

Characteristics	Requirement paragraph	Test Method
Fiber percentage		
Type I	3.3.2.1	2101
Type II	3.3.2.2	2102
Yarn ply	3.3.3	Visual 1/
Weave	3.4.2	Visual 1/
Weight	3.4.1	5041
Yarns per inch	3.4.1	5050
Breaking strength		
Warp and filling	3.4.1	5100
Shrinkage		
Warp and filling	3.4.1	5590 2/
Colorfastness to:		
Light	3.3.4.2	5660
Wet-dry cleaning	3.3.4.2	5622
Crocking	3.3.4.2	5651
Perspiration	3.3.4.2	5680
Finish	3.6	3/
pH	3.6.1	2811
Mothproofing	3.6.2	4/

1/ One determination per sample unit and the result reported as "pass" or "fail."
2/ Pressing shall be performed on a flat bed press.
3/ A certificate of compliance will be acceptable for the requirements stated.
4/ To be tested in accordance with the requirements of MIL-C-43665.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A and commercial. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A, B, and commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in dress uniforms, insignia, stripes, and piping for dress uniforms by the U. S. Marine Corps.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type of cloth required (see 1.2).

c. When a first article inspection is required (see 3.2).

d. Length required when other than specified (see 3.5).

e. Levels of packaging and packing required (see 5.1 and 5.2).

g. Color required (see 3.3.4).

* 6.2.2 Data Requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
(a) 4	Inspection System Program Plan	DI-R-4803	——
(b) 4.3	First Article Inspection Procedures	DI-T-4901	——
(c) 4.3	First Article Inspection Report	DI-T-4902	——
(d) 4.5	Production/Acceptance Inspection Procedures	DI-T-4903	——
(e) 4.5	Production Inspection Report	DI-T-4904	——

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Standard sample. For access to the standard samples, address the procuring activity issuing the invitation for bids.

6.4 First article. Examinations, tests, and approval shall be specified by the contracting officer (see 3.2).

6.5 Scales. Sears Fabric Defect Scales are available from Sears, Roebuck and Co., Department 733Q. "Sears Defect Replica Kit," Sears Tower, Chicago, IL 60684.

*6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity
Navy-MC
Project No. 8305-N013

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29405A(MC)	CLOTH, BROADCLOTH: WOOL AND WOOL/NYLON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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